Feature Selection – GLM

- **Without Normalization and Standardization –**

Feature tested: All combinations from 11 to 3 different features

'aidworkers_wounded_rate_cat', 'acled_event_rate_cat', 'cholera_cases_rate_cat', 'dep_rate', 'malaria_cases_rate_cat', 'tot_wage_cereal_smooth_lag4', 'measles_cases_rate_cat', "prop_idp", "cdi_lag5", "sam_admissions_rate_lag1", "tot_goat_cereal_smooth_lag4"

Global results - Rank

MSE	RMSE	Bias Rel	Sd RMSE	Features
487.97	20.87	-0.004796	7.611	aidworkers_wounded_rate_cat, measles_cases_rate_cat, prop_idp
492.57	20.93	-0.00490	7.777	aidworkers_wounded_rate_cat, measles_cases_rate_cat, prop_idp, cdi_lag5
758.03	26.66	-0.0092	7.247	acled_event_rate_cat, dep_rate, malaria_cases_rate_cat, prop_idp, tot_goat_cereal_smooth_lag4
642.83	24.96	-0.0057	4.67	dep_rate, malaria_cases_rate_cat, measles_cases_rate_cat, tot_goat_cereal_smooth_lag4

- **With Normalization and Standardization –**

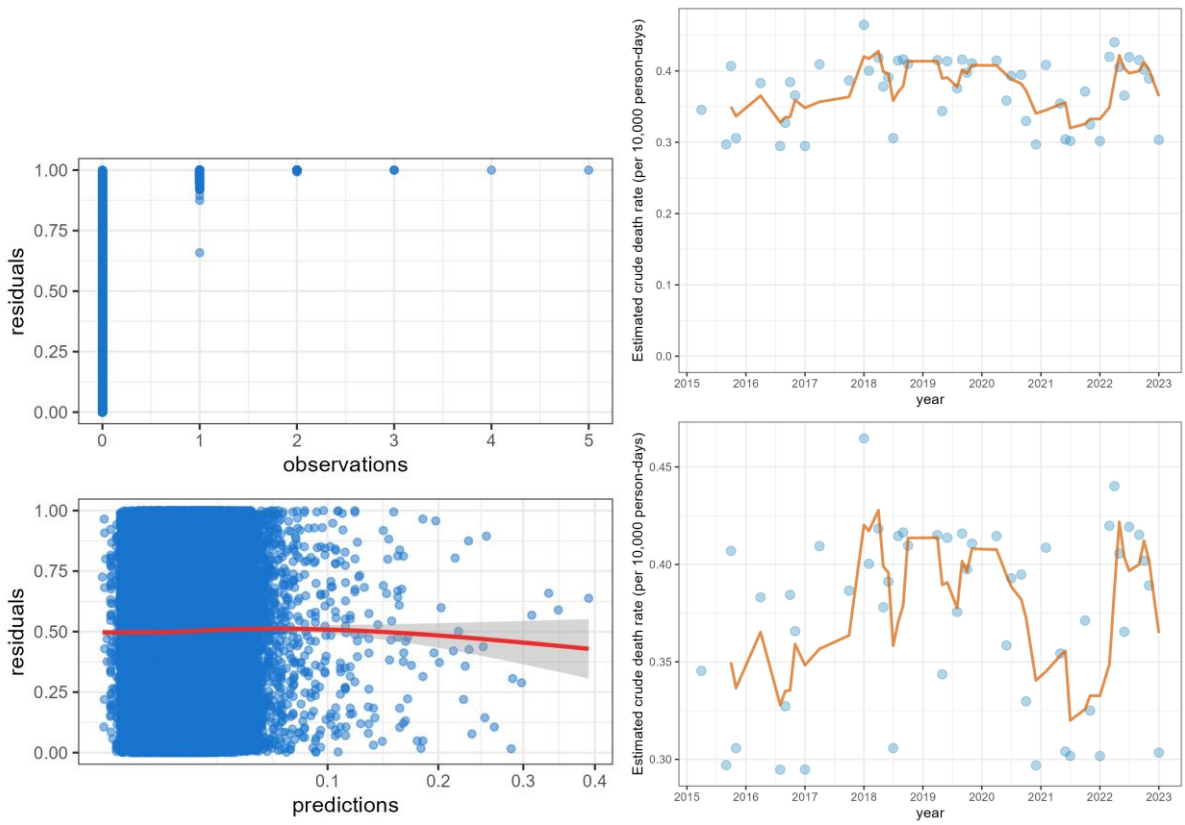
Features tested: All combinations from 11 to 3 different features

'aidworkers_wounded_rate_cat', 'acled_event_rate_cat', 'cholera_cases_rate_cat', 'dep_rate_sqt_scn', 'malaria_cases_rate_cat', 'tot_wage_cereal_smooth_lag4_scn', 'measles_cases_rate_cat', "prop_idp", "cdi_lag5", "sam_admissions_rate_lag1_scn", "tot_goat_cereal_smooth_lag4_scn"

Global results - Rank

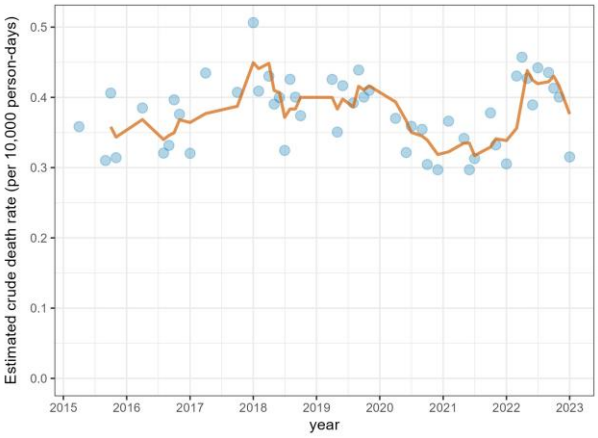
MSE	RMSE	Bias Rel	Sd RMSE	Features
487.97	20.87	-0.0048	7.61	aidworkers_wounded_rate_cat, measles_cases_rate_cat, prop_idp
492.57	20.93	-0.0049	7.777	aidworkers_wounded_rate_cat, measles_cases_rate_cat, prop_idp, cdi_lag5
607.12	23.14	-0.002	8.91	aidworkers_wounded_rate_cat, cholera_cases_rate_cat, measles_cases_rate_cat, sam_admissions_rate_lag1_scn, tot_goat_cereal_smooth_lag4_scn
668.0615	25.51387	-0.0068	4.359369	dep_rate_sqt_scn, malaria_cases_rate_cat, tot_wage_cereal_smooth_lag4_scn, measles_cases_rate_cat
686.4721	25.25577	-0.00596	7.349849	acled_event_rate_cat, dep_rate_sqt_scn, malaria_cases_rate_cat, measles_cases_rate_cat, prop_idp, sam_admissions_rate_lag1_scn

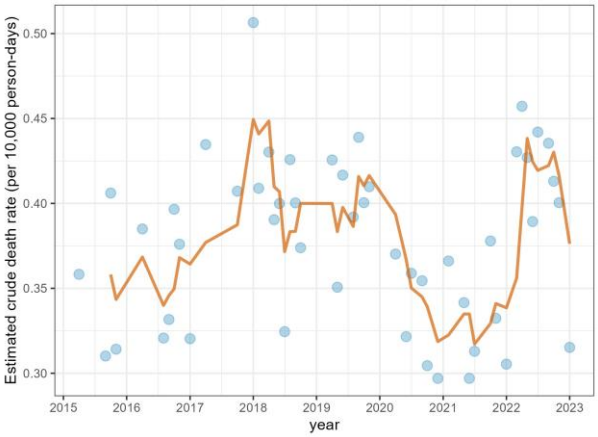
Select Feature 1- aidworkers_wounded_rate_cat, measles_cases_rate_cat, prop_idp

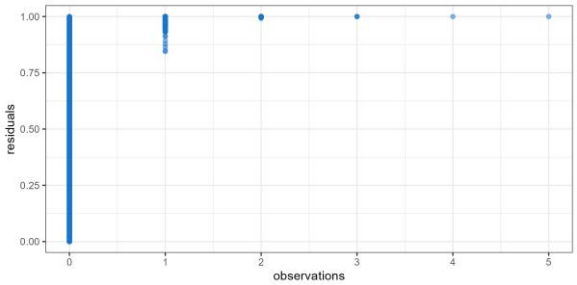


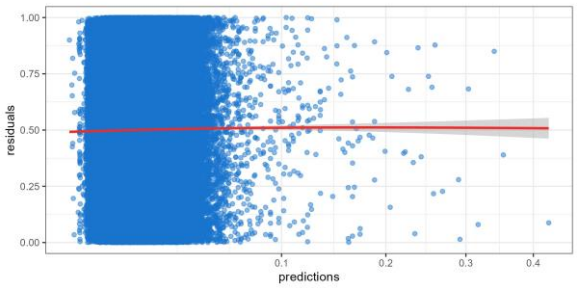
Obs	Pred	Mse	RMSE	FOLDS	BIAS_ABS	BIAS_REL
225	226.9309	3.728322	1.930886	fold0	-1.93089	-0.00858
189	211.2146	248.608	15.76731	fold1	-22.2146	-0.11754
278	247.6326	473.1324	21.75161	fold2	30.36743	0.109235
231	222.4531	373.1117	19.3161	fold3	8.546919	0.037
324	301.1026	403.3475	20.08351	fold4	22.8974	0.070671
201	247.667	699.0914	26.44034	fold5	-46.667	-0.23217
242	270.9003	718.539	26.80558	fold6	-28.9003	-0.11942
293	297.3039	631.037	25.12045	fold7	-4.30387	-0.01469
283	256.0945	641.3559	25.32501	fold8	26.90551	0.095072
251	217.7514	687.7673	26.22532	fold9	33.24861	0.132465

Select Feature 2- aidworkers_wounded_rate_cat, measles_cases_rate_cat, prop_idp, cdi_lag5



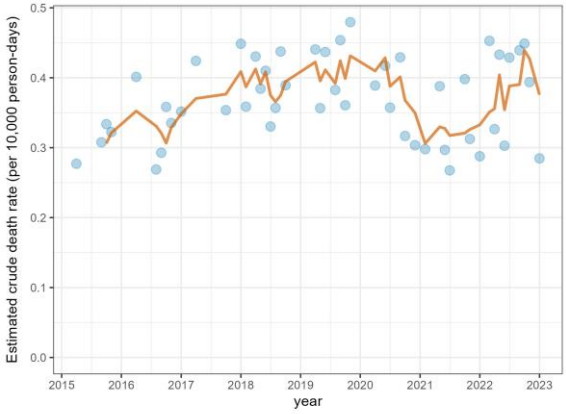


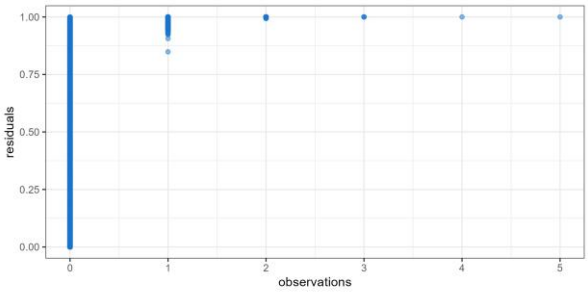


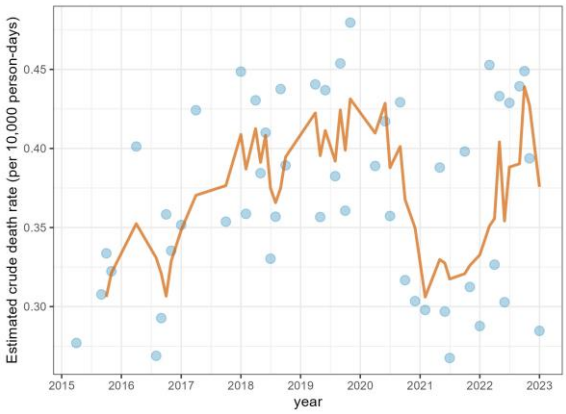


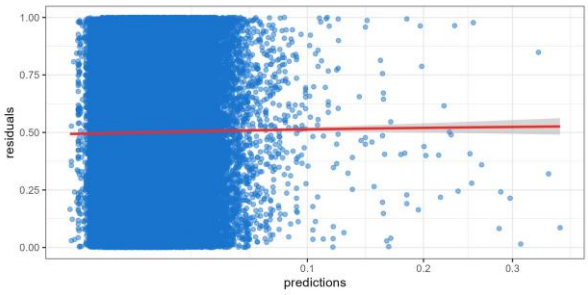
Obs	Pred	Mse	RMSE	FOLDS	BIAS_ABS	BIAS_REL
225	226.4675	2.153624	1.467523	fold0	-1.46752	-0.00652
189	210.6865	236.2282	15.36972	fold1	-21.6865	-0.11474
278	247.4117	469.3673	21.66489	fold2	30.58832	0.11003
231	216.6918	403.2069	20.08001	fold3	14.30824	0.06194
324	302.071	418.7413	20.46317	fold4	21.92895	0.067682
201	246.6497	696.2677	26.38688	fold5	-45.6497	-0.22711
242	272.5196	729.8642	27.016	fold6	-30.5196	-0.12611
293	301.5548	647.7793	25.45151	fold7	-8.55482	-0.0292
283	259.079	639.383	25.28602	fold8	23.92097	0.084526
251	218.2421	682.7527	26.12954	fold9	32.7579	0.13051

Select Feature 3- acled_event_rate_cat, dep_rate, malaria_cases_rate_cat, prop_idp, tot_goat_cereal_smooth_lag4



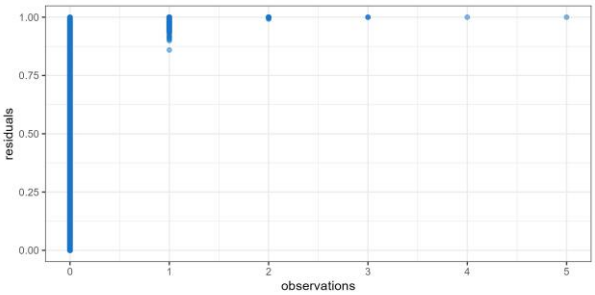


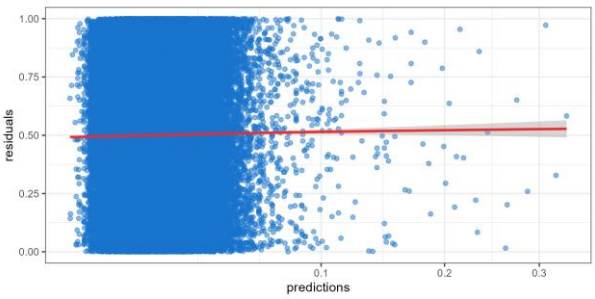


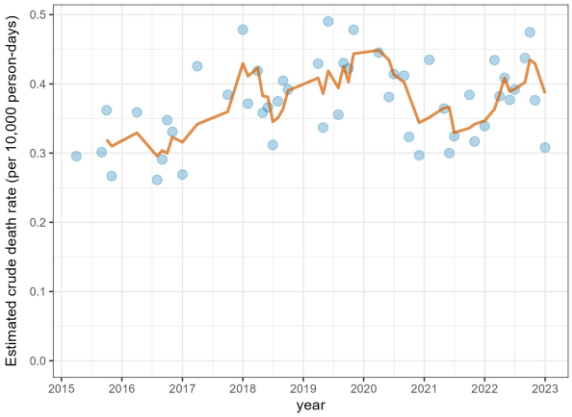


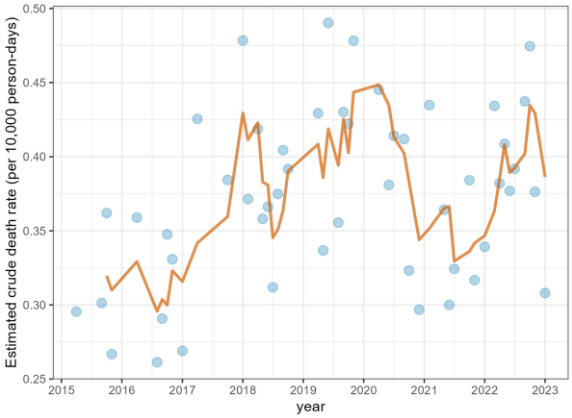
Obs	Pred	Mse	RMSE	FOLDS	BIAS_ABS	BIAS_REL
225	234.6708	93.52468	9.670816	fold0	-9.67082	-0.04298
189	216.4013	422.1779	20.54697	fold1	-27.4013	-0.14498
278	236.6559	851.2312	29.17587	fold2	41.34414	0.14872
231	229.0776	639.3473	25.28532	fold3	1.922428	0.008322
324	301.5813	611.9975	24.73858	fold4	22.41871	0.069194
201	250.0258	910.5853	30.17591	fold5	-49.0258	-0.24391
242	262.003	837.6615	28.94238	fold6	-20.003	-0.08266
293	307.0297	757.5578	27.52377	fold7	-14.0297	-0.04788
283	245.5655	829.0895	28.79391	fold8	37.43452	0.132277
251	218.9544	848.8727	29.13542	fold9	32.04562	0.127672

Select Feature 4- aidworkers_wounded_rate_cat, cholera_cases_rate_cat, measles_cases_rate_cat, sam_admissions_rate_lag1_scn, tot_goat_cereal_smooth_lag4_scn



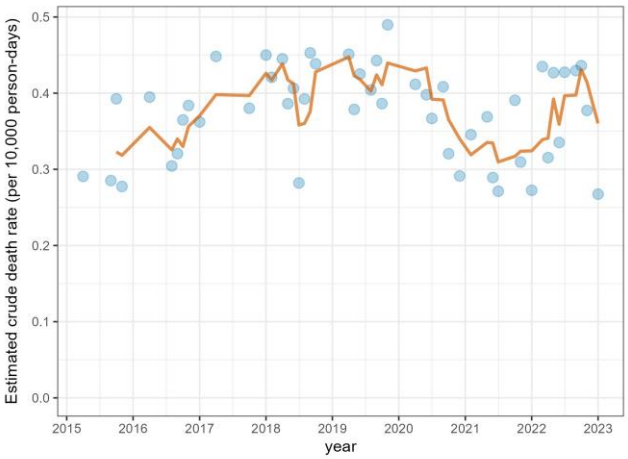


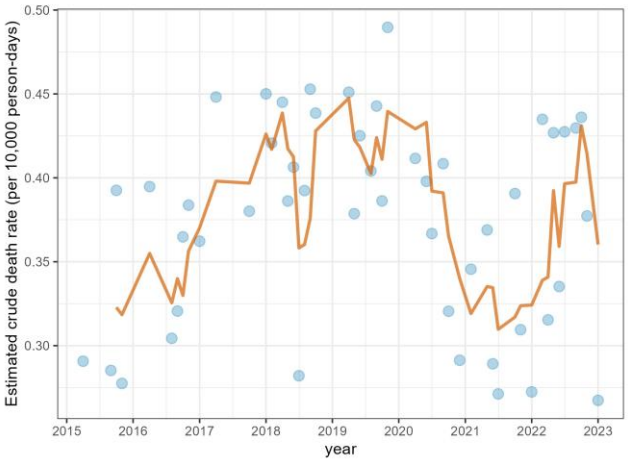


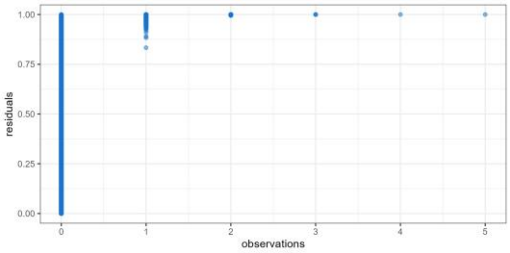


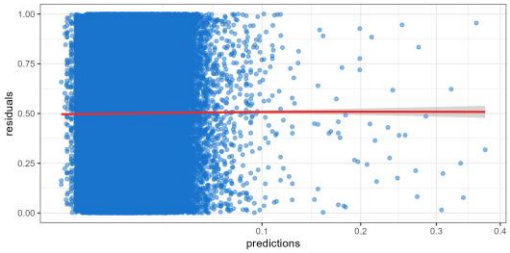
Obs	Pred	Mse	RMSE	FOLDS	BIAS_ABS	BIAS_REL
225	222.0403	8.759633	2.959668	fold0	2.959668	0.013154
189	205.8551	146.4274	12.10072	fold1	-16.8551	-0.08918
278	238.2541	624.1979	24.98395	fold2	39.74593	0.142971
231	214.476	536.4095	23.16051	fold3	16.52405	0.071533
324	297.3454	571.221	23.90023	fold4	26.65459	0.082267
201	252.5723	919.302	30.31999	fold5	-51.5723	-0.25658
242	277.4152	967.1495	31.09903	fold6	-35.4152	-0.14634
293	304.8216	863.7247	29.38919	fold7	-11.8216	-0.04035
283	263.3361	810.7185	28.47312	fold8	19.66389	0.069484
251	220.2061	824.4731	28.71364	fold9	30.79391	0.122685

Select Feature 5- dep_rate, malaria_cases_rate_cat, measles_cases_rate_cat, tot_goat_cereal_smooth_lag4



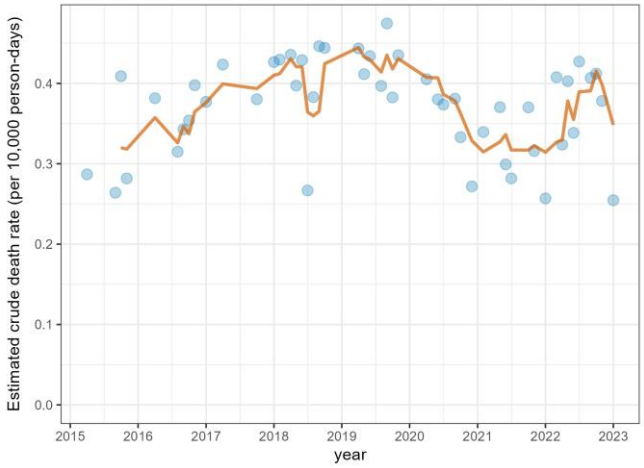


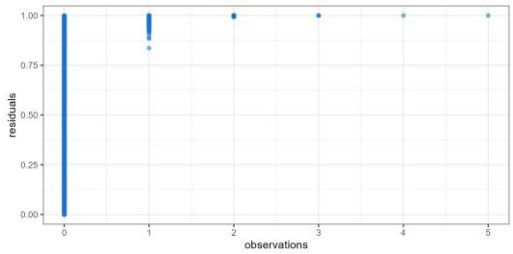


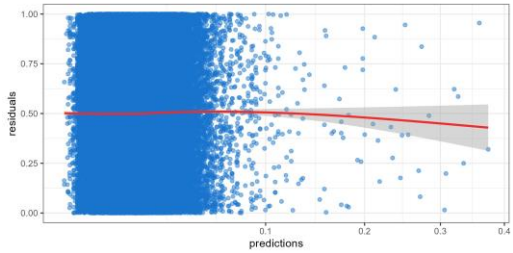


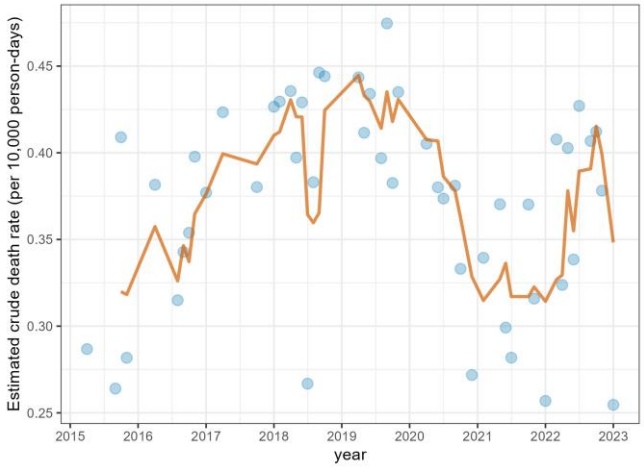
Obs	Pred	Mse	RMSE	FOLDS	BIAS_ABS	BIAS_REL
225	231.8754	47.27178	6.875447	fold0	-6.87545	-0.03056
189	213.5887	325.9384	18.05376	fold1	-24.5887	-0.1301
278	247.198	533.5474	23.09865	fold2	30.80204	0.110799
231	218.6328	438.3973	20.93794	fold3	12.36717	0.053538
324	301.4384	452.5229	21.27259	fold4	22.56159	0.069635
201	250.1353	779.4823	27.91921	fold5	-49.1353	-0.24445
242	270.2651	782.2582	27.96888	fold6	-28.2651	-0.1168
293	296.9074	686.3844	26.19894	fold7	-3.90742	-0.01334
283	249.7173	733.2016	27.0777	fold8	33.28271	0.117607
251	217.4366	772.5316	27.79445	fold9	33.5634	0.133719

Select Feature 6- dep_rate_sqt_scn, malaria_cases_rate_cat, tot_wage_cereal_smooth_lag4_scn, measles_cases_rate_cat



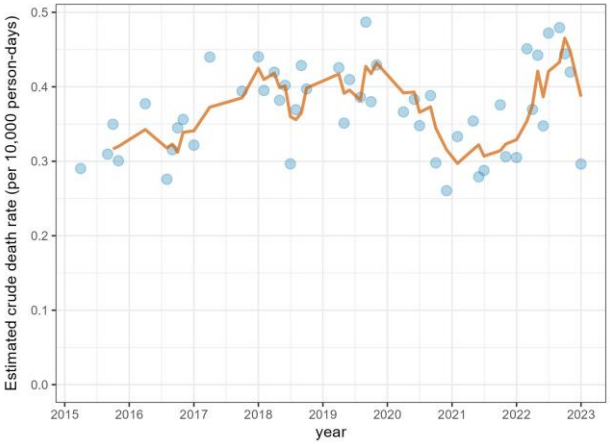


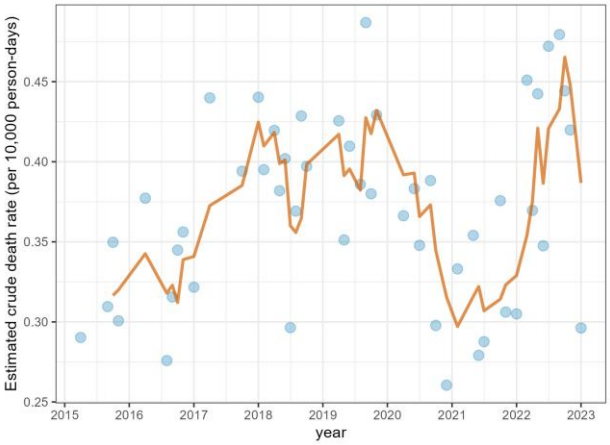


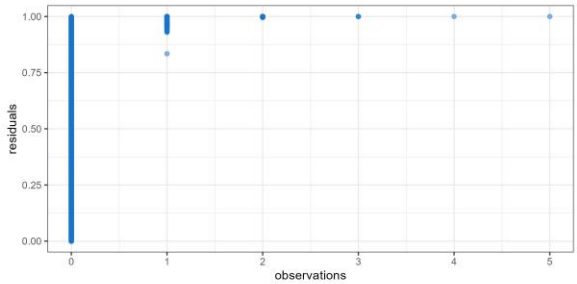


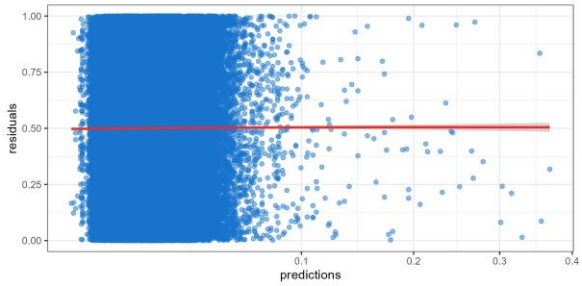
Obs	Pred	Mse	RMSE	FOLDS	BIAS_ABS	BIAS_REL
225	240.3368	235.2183	15.33683	fold0	-15.3368	-0.06816
189	215.5878	471.0646	21.70402	fold1	-26.5878	-0.14068
278	240.5271	782.1167	27.96635	fold2	37.47294	0.134795
231	225.0054	595.5714	24.40433	fold3	5.99463	0.025951
324	300.8447	583.6911	24.1597	fold4	23.15534	0.071467
201	250.0343	887.136	29.78483	fold5	-49.0343	-0.24395
242	261.4093	814.2194	28.53453	fold6	-19.4093	-0.0802
293	297.8244	715.3513	26.74605	fold7	-4.8244	-0.01647
283	245.0075	796.2487	28.21788	fold8	37.99247	0.134249
251	222.1255	799.9973	28.28422	fold9	28.87447	0.115038

Select Feature 7- acled_event_rate_cat, dep_rate_sqt_scn, malaria_cases_rate_cat, measles_cases_rate_cat, prop_idp, sam_admissions_rate_lag1_scn









Obs	Pred	Mse	RMSE	FOLDS	BIAS_ABS	BIAS_REL
225	233.1544	66.49425	8.154401	fold0	-8.1544	-0.03624
189	210.8141	271.1744	16.46737	fold1	-21.8141	-0.11542
278	230.3239	938.4515	30.63416	fold2	47.67605	0.171497
231	216.8104	754.1748	27.46224	fold3	14.18959	0.061427
324	300.9771	709.3504	26.63363	fold4	23.02288	0.071058
201	246.9528	943.0687	30.70942	fold5	-45.9528	-0.22862
242	264.376	879.8712	29.66262	fold6	-22.376	-0.09246
293	313.7483	823.699	28.70016	fold7	-20.7483	-0.07081
283	261.2061	784.9517	28.01699	fold8	21.79389	0.07701
251	223.1652	783.9344	27.99883	fold9	27.83484	0.110896